KAUST Thuwal Campus Finals, 3-Seat Relay & Academy Synthesis

ClipSAT Printable Practice Suite (clipsat.org) • Topic 12.01 • KAUST (■■■■■■■ ■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■)

ANSWER KEY — COMPLETE WORKED SOLUTIONS

KAUST Thuwal Campus Finals, 3-Seat Relay & Academy Synthesis Practice Exercises

1 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/5}$ be a primitive 5-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{4} (2 - \zeta^k)$.

a. 32

b. 17

c. 31 ✓

d. 27

Because $1, \zeta, \zeta^2, \ldots, \zeta^4$ are all 5 roots of $z^5 - 1$, dividing out $(z - 1)$ yields the cyclotomic identity $\prod_{k=1}^{4} (z - \zeta^k) = \frac{z^5 - 1}{z - 1} = 1 + z + \ldots + z^4$. Substituting $z = 2$ gives 31.

2 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/6}$ be a primitive 6-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{5} (2 - \zeta^k)$.

a. 64

b. 63 ✓

c. 33

d. 59

Because $1, \zeta, \zeta^2, \ldots, \zeta^5$ are all 6 roots of $z^6 - 1$, dividing out $(z - 1)$ yields the cyclotomic identity $\prod_{k=1}^{5} (z - \zeta^k) = \frac{z^6 - 1}{z - 1} = 1 + z + \ldots + z^5$. Substituting $z = 2$ gives 63.

3 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/7}$ be a primitive 7-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{6} (2 - \zeta^k)$.

a. 127 ✓

b. 128

c. 65

d. 123

Because $1, \zeta, \zeta^2, \ldots, \zeta^6$ are all 7 roots of $z^7 - 1$, dividing out $(z - 1)$ yields the cyclotomic identity $\prod_{k=1}^{6} (z - \zeta^k) = \frac{z^7 - 1}{z - 1} = 1 + z + \ldots + z^6$. Substituting $z = 2$ gives 127.

© ClipSAT — clipsat.org | All Rights Reserved

4 [KAUST Thuwal Finals] Let $\zeta = e^{2\pi i/8}$ be a primitive 8-th root of unity (plotted on the Argand diagram). Evaluate the exact complex product $\prod_{k=1}^{7} (2 - \zeta^k)$.

a. 256

b. 129

c. 251

d. 255 ✓

Because $1, \zeta, \zeta^2, \ldots, \zeta^7$ are all 8 roots of $z^8 - 1$, dividing out $(z-1)$ yields the cyclotomic identity $\prod_{k=1}^{7} (z - \zeta^k) = \frac{z^8-1}{z-1} = 1 + z + \ldots + z^7$. Substituting $z = 2$ gives 255.

5 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 6 \\ j \equiv 0 \pmod 3}} \binom{6}{j}$.

a. 21

b. 24

c. 22 ✓

d. 16

Let $P(x) = (1+x)^6$. By the Roots-of-Unity Filter, $\sum_{j \equiv 0 \pmod 3} \binom{6}{j} = \frac{1}{3}(P(1) + \omega^{-0}P(\omega) + \omega^{-0}P(\omega^2))$. Using $1 + \omega = -\omega^2$ and $1 + \omega^2 = -\omega$, the sum simplifies to $\frac{1}{3}\left(2^6 + 2\cos\frac{(6-0)\pi}{3}\right) = 22$.

6 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 6 \\ j \equiv 1 \pmod 3}} \binom{6}{j}$.

a. 23

b. 21 ✓

c. 16

d. 22

Let $P(x) = (1+x)^6$. By the Roots-of-Unity Filter, $\sum_{j \equiv 1 \pmod 3} \binom{6}{j} = \frac{1}{3}(P(1) + \omega^{-1}P(\omega) + \omega^{-2}P(\omega^2))$. Using $1 + \omega = -\omega^2$ and $1 + \omega^2 = -\omega$, the sum simplifies to $\frac{1}{3}\left(2^6 + 2\cos\frac{(6-2)\pi}{3}\right) = 21$.

7 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 7 \\ j \equiv 2 \pmod 3}} \binom{7}{j}$.

a. 42 ✓

b. 44

c. 32

d. 43

Let $P(x) = (1+x)^7$. By the Roots-of-Unity Filter, $\sum_{j \equiv 2 \pmod 3} \binom{7}{j} = \frac{1}{3}(P(1) + \omega^{-2}P(\omega) + \omega^{-4}P(\omega^2))$. Using $1 + \omega = -\omega^2$ and $1 + \omega^2 = -\omega$, the sum simplifies to $\frac{1}{3}\left(2^7 + 2\cos\frac{(7-4)\pi}{3}\right) = 42$.

© ClipSAT — clipsat.org | All Rights Reserved

8 [KAUST Academy] Applying the Roots-of-Unity Filter with $\omega = e^{2\pi i/3}$ (shown on the unit circle), compute the exact stepped binomial sum $\sum_{\substack{0 \le j \le 7 \\ j \equiv 0 \pmod 3}} \binom{7}{j}$.

a. 42

b. 45

c. 32

d. 43 ✓

Let $P(x) = (1+x)^7$. By the Roots-of-Unity Filter, $\sum_{j \equiv 0 \pmod 3} \binom{7}{j} = \frac{1}{3}(P(1) + \omega^{-0}P(\omega) + \omega^{-0}P(\omega^2))$. Using $1 + \omega = -\omega^2$ and $1 + \omega^2 = -\omega$, the sum simplifies to $\frac{1}{3}\left(2^7 + 2\cos\frac{(7-0)\pi}{3}\right) = 43$.

9 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{10} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^2)(1-x^5)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 2v + 5w = 10$.

a. 12

b. 8

c. 10 ✓

d. 14

Expanding each geometric factor $(1-x^d)^{-1} = \sum_{k \ge 0} x^{kd}$, $[x^{10}]G(x)$ counts non-negative triples (u, v, w) with $1u + 2v + 5w = 10$. Case-splitting on $w = 0, 1, \ldots, \lfloor 10/5 \rfloor$ and summing $\#\{(u, v) : 1u + 2v = 10 - 5w\}$ yields exactly 10.

10 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{11} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^2)(1-x^3)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 2v + 3w = 11$.

a. 18

b. 16 ✓

c. 14

d. 20

Expanding each geometric factor $(1-x^d)^{-1} = \sum_{k \ge 0} x^{kd}$, $[x^{11}]G(x)$ counts non-negative triples (u, v, w) with $1u + 2v + 3w = 11$. Case-splitting on $w = 0, 1, \ldots, \lfloor 11/3 \rfloor$ and summing $\#\{(u, v) : 1u + 2v = 11 - 3w\}$ yields exactly 16.

© ClipSAT — clipsat.org | All Rights Reserved

11 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{12} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^3)(1-x^5)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 3v + 5w = 12$.

a. 9 ✓

b. 11

c. 7

d. 13

Expanding each geometric factor $(1 - x^d)^{-1} = \sum_{k \geq 0} x^{kd}$, $[x^{12}]G(x)$ counts non-negative triples (u, v, w) with $1u + 3v + 5w = 12$. Case-splitting on $w = 0, 1, \ldots, \lfloor 12/5 \rfloor$ and summing $\#\{(u, v) : 1u + 3v = 12 - 5w\}$ yields exactly 9.

12 [KAUST Academy — Cambridge Summer Prep] Find the coefficient of x^{13} in the formal power series expansion of $G(x) = \frac{1}{(1-x^1)(1-x^2)(1-x^4)}$, i.e., the number of non-negative integer solutions (u, v, w) to $1u + 2v + 4w = 13$.

a. 18

b. 14

c. 20

d. 16 ✓

Expanding each geometric factor $(1 - x^d)^{-1} = \sum_{k \geq 0} x^{kd}$, $[x^{13}]G(x)$ counts non-negative triples (u, v, w) with $1u + 2v + 4w = 13$. Case-splitting on $w = 0, 1, \ldots, \lfloor 13/4 \rfloor$ and summing $\#\{(u, v) : 1u + 2v = 13 - 4w\}$ yields exactly 16.

© ClipSAT — clipsat.org | All Rights Reserved